# Extended Protocol

Extended protocol detail for analyses summarised in the main text and Supplementary Results. Nothing here changes an estimand, a count or a reported result; these sections record the selection rules, calibration protocols and sensitivity estimators in full.

## M1. Edge-level holdout: full procedure

To test reproducibility at the level of the individual edge we used the atlas's donor-pair and per-guide differential expression. The `by_donors` object contains one differential-expression fit per pair of donors; with four donors this gives six pairs and three ways to split them into two disjoint, non-overlapping donor pairs. For each split we defined the reversing edges in one pair (by the census criterion: significant at adjusted $p < 0.10$ in both Rest and Stim48hr, opposite sign) and, in the disjoint pair, asked whether each reversal recurred, requiring the edge to be significant in both states in the held-out pair and to reproduce both its Rest sign and its Stim48hr sign. We ran all six train/test directions. To avoid pseudo-replication, reversing edges were deduplicated to unique (regulator, target) pairs (a pair may be testable in more than one split), and a unique edge was scored as reproduced if it reproduced in the majority of the folds in which it was testable; reproduction rates are reported within cumulative bins of the training minimum absolute log2 fold change, with 95% confidence intervals obtained by bootstrapping over the contributing regulators rather than over edges, since edges sharing a regulator share a perturbation and a cell population and are not independent (an edge-level Wilson interval [@wilson1927probable] is roughly a third too narrow on the headline bin, and on the four-regulator guide split is spuriously informative where the clustered interval spans the whole range), alongside the same-sign reproduction rate of stable concordant edges as a ceiling. Significance was assessed by a within-regulator target-label permutation (2,000 iterations): within each fold and regulator the test-fold sign-pairs were permuted across that regulator's testable targets, so the null preserves the regulator's sign composition and asks whether the specific train-to-test edge correspondence reproduces more than random target matching. Because a regulator's testable targets are overwhelmingly same-sign, this null is 2.3% in the headline bin. The analogous `by_guide` procedure (define in `guide_1`, test in `guide_2`, and reverse) is a guide-consistency sensitivity. Eight unique edges across four regulator clusters are testable, so it cannot rule out guide-specific or shared technical effects. Because donor-pair and single-guide fits are far less powered than the pooled analysis, only reversals significant in both states in both folds are testable; the holdout therefore evaluates a high-confidence subset.

The donor-pair and per-guide differential-expression objects used here are the atlas releases `GWCD4i.DE_stats.by_donors.h5mu` and `GWCD4i.DE_stats.by_guide.h5mu`.

## M2. Per-regulator enrichment: full calibration protocol

We chose these tests over a covariate-adjusted logistic regression on the basis of calibration rather than on principle. For the primary Fisher test, each null census held every regulator's out-degree and the total number of reversals within each degree stratum fixed, then allocated those reversals across regulators by a multivariate hypergeometric draw. Across 2,000 null censuses, the two-sided Fisher plus BH procedure made any false discovery in 1.85% of replicates, with 0.019 calls on average and at most two; the one-sided enrichment sensitivity gave 1.65%. The implementation of the two-sided probability ordering agreed with SciPy's `fisher_exact` to $1.11 \times 10^{-15}$. The number of two-sided calls varied from 91 to 112 across 5, 8, 10, 12, 15 and 20 degree strata, and 16–18 of the 20 smallest-$p$ regulators under the ten-stratum specification were retained at every alternative resolution, so the broad result is stable but the exact ranked set is partly bin-sensitive. By contrast, the complex-level permutation false-positive rate ranged from 2.0% to 12.5% over 200 null replicates for the four testable complexes, while CBP/p300 and PAF1C had no testable null replicates under the matching rule. Complex-level $p$-values therefore carry no confirmatory interpretation. A logistic regression of reversal on membership with $\log_{10}$ out-degree as a covariate and standard errors clustered by regulator also failed these checks, calling 236 and 229 of 264 regulators significant on null data and reaching false-positive rates of 0.18 to 0.85 at the complex level. The failure is expected because the membership indicator is constant within a regulator and a single-gene test has exactly one treated cluster. Logistic estimates are retained only in a column explicitly marked uncalibrated and annotate no figure. Calibration code and outputs are `fig2a_enrichment_calibration.py`, `fig2a_enrichment_calibration.json`, `enrichment_test_calibration.py` and `enrichment_test_calibration.json`.

Out-degree predicts reversal rate (Spearman $\rho = 0.235$, $p = 1.2 \times 10^{-4}$ across the 264 regulators with at least 50 comparable trans edges), which is why the per-regulator test conditions on it. For regulator $i$ in stratum $s$, we formed the table $[[k_i, n_i - k_i], [K_s - k_i, (N_s - K_s) - (n_i - k_i)]]$, where $n_i$ and $k_i$ are that regulator's both-significant and reversing trans edges and $N_s$ and $K_s$ are the corresponding stratum totals. A two-sided Fisher exact test therefore conditions on the stratum reversal total and compares the regulator with all other edges in its stratum without treating an estimated rate as fixed. For the architecture analysis, each regulator's ratio to its degree-matched expectation was computed over the identity-safe Rest-versus-Stim48 master catalogue.

## M3. Target-side analyses: external audit of the direction test

The disjoint donor-pair and Arce tables were audited at one record per regulator-target pair within each directional fold or assay, requiring the edge to reverse in the validation data. Reciprocal folds and Arce assays were not pooled. Only 7–17 qualifying targets remained per donor direction, two in Arce Perturb-seq and none in the Arce bulk assay, so these datasets were underpowered for a confirmatory target-direction test and no external replication result was reported.

Sign concordance was recomputed within each decile of target baseline expression and target in-degree to test robustness to those measured features (Supplementary Figure S2b). For the target-conditioned direction-unanimity test, only targets retaining at least one non-reversing edge could contribute. The primary threshold was at least two reversals per target; thresholds of three to five, leave-one-regulator-out over the five largest contributors and removal of CCNC, AHR and ARNT together were sensitivity analyses. A separate within-regulator direction permutation was rerun on the same eligible targets as a complementary sensitivity.

## M4. Target-activation trajectory: regulator-margin sensitivity

As a separate sensitivity, reversal labels were uniformly reassigned within each regulator-by-Rest-sign fibre while holding the observed number of positive-to-negative and negative-to-positive reversals for each regulator fixed. The eligibility rule of at least two reversals and at least one stable edge was reapplied in every draw; target membership and target reversal counts could therefore vary. We used 10,000 draws and the same one-sided plus-one rule. This randomisation preserves regulator signed-reversal margins, eligible dyads and structural zeros but not target reversal counts; the target-conditioned randomisation preserves the complementary target margins but not regulator burdens. We report the two nulls separately and do not interpret their agreement as a simultaneous fixed-margin test, a causal decomposition or a percentage of target control. The non-unanimous subset was examined descriptively without a $p$-value because membership is defined by the observed reversal pattern.

## M5. Target-activation trajectory: same-laboratory Arce support

For same-laboratory support, we scored the frozen unanimous directions in non-targeting-control Arce pseudobulk counts separately for Teff and Treg. Counts were converted to log2(CPM+1), and the response was Stimulated minus Resting within each of donors A and B. Targets were retained at mean CPM at least 5 in the relevant cell context. We reported an unstratified target-level Fisher test and, separately, an exact fixed-margin permutation stratified by expression quartile, census in-degree quartile and reversal-count stratum; donor AUCs were reported separately. With only two donors, these comparisons are descriptive and provide neither donor-population inference nor Teff specificity.

## M6. Promoter-accessibility compatibility: full design

We froze the GSE116696 design before accessing target-linked ATAC outcomes [@yukawa2020ap1]. The deposited processed files are per-library MACS2 island lists rather than a common normalised count matrix. The primary comparison used three naive and three 5 h CD3/CD28-stimulated naive-CD4 libraries. GENCODE v19 transcription start sites were mapped to hg19 promoter windows of $\pm 2$ kb, and each target-library value was binary island presence. The target outcome was mean presence across the three stimulated libraries minus mean presence across the three naive libraries; $\pm 1$ kb and $\pm 5$ kb were descriptive window sensitivities. The frozen target score and 903-target panel were taken unchanged from the GSE140244 specificity analysis.

The primary statistic was Spearman correlation between signed reversal score and the 5 h promoter outcome. The target-conditioned null used 100,000 exact within-target hypergeometric reassignments, preserving target identity, in-degree, reversal count and complete Rest-sign composition. A separate 10,000-draw regulator-margin sensitivity reassigned reversal labels across the full 83,489-edge identity-safe fibre within regulator-by-Rest-sign strata, then reapplied the target eligibility rule. Chromosomes were the resampling blocks for the effect interval. CD28-dependence and A-FOS contrasts formed a two-test Holm family and had predeclared information gates based on the number and fraction of non-zero target outcomes. Direct uniform samplers were used; no Markov chain was required. Because the processed islands lack a shared quantitative matrix and donor correspondence, this analysis tests binary promoter compatibility only, not quantitative accessibility, distal enhancer linkage or mechanism.

## M7. Kinetics: resolved 8 h sign states and functional annotation

Timing was assessed conditionally among reversals with an 8 h adjusted $p < 0.10$; all others were retained as unresolved. A resolved edge was Rest-sign-retaining when its 8 h coefficient matched its resting sign and Stim48-sign-matching when it matched its 48 h sign. Rest-sign-retaining fractions were calculated separately for lost and gained effects and compared within the 75 regulators containing at least one resolved edge of each orientation using a two-sided paired Wilcoxon signed-rank test; exact fractional differences defined ties, and zero differences were excluded by the standard Wilcoxon zero method. Separately, the per-regulator Rest-sign-retaining fraction was calculated for the 50 regulators with at least five resolved reversals. For the functional analysis, gene-set membership was taken from external libraries, Gene Ontology Biological Process 2023 [@ashburner2000go; @geneontology2023knowledgebase] and CORUM [@steinkamp2025corum], accessed via Enrichr [@chen2013enrichr; @kuleshov2016enrichr; @xie2021enrichr], rather than curated here, since the regulator sign states were known by that point and curating sets afterwards would be circular; the background was the tested regulators themselves rather than the genome, because the 617 represented census regulators are already heavily enriched for chromatin and transcription factors by construction and a genome background would return that selection as biology. Annotation density was quantified as the number of Gene Ontology Biological Process terms containing each regulator and correlated against the Rest-sign-retaining fraction. The sign rule was written after the census existed and is an organisational description of already-defined reversals, not a selection rule; no edge is defined or excluded by its 8 h value. Three controls accompany it. Resolved and unresolved edges were compared on their Rest-to-48 h span and 8 h standard error by Mann-Whitney test, to check that the unresolved majority is not a power class. The same sign rule was applied to the stable edges resolved at 8 h, whose two endpoints share a sign and therefore have a known right answer, as a calibration control. Finally, because a stable edge with one poorly estimated endpoint would enter the census as a reversal and read as Rest-sign-retaining, we compared $\log(\mathrm{se}_{48}/\mathrm{se}_{Rest})$ between the two resolved classes; that artefact predicts the larger value for Rest-sign-retaining edges. Values are written by `scripts/eight_hour_switch_timing.py` to `results_data/census/eight_hour_switch_timing.json`.

Two design properties follow from the geometry of a reversing edge. First, an edge caught mid-transit has an 8 h effect passing through zero, so all coherence analyses use the continuous z-statistic (log2 fold change divided by its standard error) supplied in the released `zscore` layer. Second, the position of the 8 h effect along the Rest-to-Stim48hr interval, $t = (\mathrm{lfc}_8 - \mathrm{lfc}_{\mathrm{Rest}})/(\mathrm{lfc}_{\mathrm{Stim48}} - \mathrm{lfc}_{\mathrm{Rest}})$, is uninformative here: because a reversing edge runs between effects of opposite sign, an 8 h effect of zero yields $t \approx 0.5$ regardless.

In the identity-safe join to Stim8hr statistics, KAT7's two reversals account for the reversal and high-magnitude-core differences.

Coherence was assessed per regulator over regulators with at least eight reversing edges (110 regulators, covering 88.3% of the kinetic reversal subset). The null permutes the 8 h values across a regulator's edges ($B = 2{,}000$), breaking target identity while preserving that regulator's 8 h distribution and its endpoint sign composition. As a robustness check we repeated the analysis on the 3,119 reversals exceeding an absolute z of 2.5 in both defining states.

## M8. Donor-balanced decomposition: unit eligibility and donor support

The response-shape analysis was frozen before target-expression access and retained the full 4,379-edge census as its reporting denominator. The biological replicate was donor. For each regulator, the first two complete identity-consistent physical guides in library order were fixed as guides A and B. A donor–regulator–condition unit required at least five clean cells for each guide, at least 30 across A and B, at least one contributing lane with five combined case cells and same-lane reference and CAL reservoirs containing at least 20 physical guides and 50 cells per role. Donors had to pass in both Rest and Stim48hr, and metadata eligibility required at least three paired donors. Case and control summaries were formed within donor, condition and lane; physical reference guides were averaged equally within lane, lanes were combined using each case guide's frozen case-cell weights, guides A and B were weighted equally and the same donor identities were averaged equally across conditions.

Guide-level detection probability $p_g$ and all-cell marginal mass $q_g$ were computed first; regulator-level values were $p = (p_A + p_B)/2$, $q = (q_A + q_B)/2$ and $m = q/p$. We did not directly average per-guide positive-cell means. A common donor entered an edge only when, in both conditions, each case guide contained at least ten target-detected cells and each separately guide-weighted reference arm contained at least ten raw target detections represented across at least ten distinct physical reference guides. At least three identical common donors were required across both guides and conditions. Unsupported rows remained in the census as `UNANALYSABLE_METADATA_SUPPORT` or `UNANALYSABLE_POSITIVE_INTENSITY`.

The primary target value was target-excluded CP10K, defined for target $t$ in cell $i$ as $10{,}000 Y_{it}/(L_i - Y_{it})$, where $Y_{it}$ is raw target UMI and $L_i$ is total UMI; detection was defined as $Y_{it} > 0$. For arm $a$, $M_a = p_a m_a$ is the all-cell marginal mean.

For every metric $x$, the cross-condition change was $\Delta_x = x_{Stim48hr} - x_{Rest}$, and the orientation $o$ was fixed to the census sign expected in Stim48hr ($o = +1$ for effects gained on stimulation, $o = -1$ for effects lost), so oriented changes $o\Delta_x$ are positive when they follow the frozen reversal direction. CAL and regulator-matched stable comparisons inherited their reversal parent's $o$; the stable edge's observed same-sign direction was selection provenance only. In the dominant-component comparison among the 1,899 raw-compatible reversals, no exact ties occurred.

Physical-guide agreement compared the signs of the separately reconstructed equal-donor $o\Delta_T$ values for guides A and B across all 1,910 continuous estimates. Leave-one-donor-out agreement used the same set restricted to edges whose frozen common set contained all four donors: for each omitted donor we recomputed the equal-donor Stim48hr-minus-Rest total change over the remaining three, oriented it by $o$ and compared its sign with the full four-donor value.

## M9. Donor-balanced decomposition: calibration-guide (CAL) selection and envelopes

CAL pseudo-perturbations were fixed before target access. Each physical guide was represented by its clean-cell counts over the 190 donor–condition–lane blocks, with absent blocks set to zero. For the regulator's exact paired-donor set, a candidate CAL guide required at least five cells in both conditions for every donor; an eligible ordered pair of distinct CAL guides required at least 30 combined cells in every donor–condition and, in every required donor–condition, at least five combined cells in one same-lane block passing the 20-guide and 50-cell-per-role REF/CAL reservoir gates. Eligible pairs were scored by $d(A, c_A) + d(B, c_B)$, where $d(g, c)$ was the mean over blocks of $|\log(1 + n_g) - \log(1 + n_c)|$; the minimum score, then lexicographic guide IDs, fixed one pair. CAL never entered REF. There was no CAL-guide reuse cap and there were zero rematching draws. The pair was reused across all donors, conditions, reversal targets and stable targets for that regulator and passed the same estimator. For each metric and condition, and for Stim48hr minus Rest, we calculated one global absolute 95th sample percentile by Hyndman–Fan type 7 over equally weighted, unique supported reversal-parent CAL pseudo-edges; an envelope was available only with at least 1,000 such pseudo-edges. All envelopes were supported by 1,726 unique CAL parents; CAL pseudo-edges attached to stable targets were excluded.

## M10. Donor-balanced decomposition: regulator-matched stable comparison family

Stable biological context used only the pre-outcome regulator-matched family. Stable edges were matched exactly to the parent reversal on regulator Ensembl identity and Rest sign, with up to two controls per parent selected by coverage-first capacitated minimum-cost matching. Euclidean distance was computed after robust scaling of Rest and Stim48hr baseline-expression deciles, target in-degree, target testability and weaker-endpoint magnitude; a physical stable edge could be reused by at most five parents. Target-level per-cell outcomes were not used in matching. The response-shape support pipeline retained 3,905 parent-specific regulator-matched comparison rows, representing 2,874 unique physical stable regulator–target edges and 1,965 unique stable target genes; these supplied descriptive biological context and never entered CAL thresholds.

## M11. Donor-balanced decomposition: raw target-UMI-rate sensitivity

For the raw target-UMI-rate sensitivity, within arm $a$, donor $d$, condition $c$ and lane $l$, $K_{adcl}$ was the summed raw target UMI and $E_{adcl}$ the summed total UMI after subtracting target UMI. We defined $r_{adcl} = (K_{adcl} + 0.5)/E_{adcl}$. For each case guide, $\log_2(r_{case}/r_{REF})$ was calculated separately against every physical reference guide in the same lane; reference comparisons were averaged equally within lane, lanes were combined with the frozen case-cell weights, guides A and B were averaged equally, and exactly the primary common donors were averaged equally. CAL used the identical estimator. An effect was labelled `MIXED_OR_UNRESOLVED` only when, in either condition, the absolute donor-averaged log2 rate ratio was at least 0.15, strictly exceeded the condition-specific raw-rate CAL envelope and had the opposite non-zero sign to the marginal CP10K effect. This veto affected 11 of the 1,910 continuous estimates. Continuous estimates were reported without individual-edge p-values, cell-level tests or class-enrichment inference.

## M12. Census construction: on-target self-effect audit and plug-in sign-flip sensitivity

On-target self effects were identified by stable Ensembl identity rather than by displayed gene symbol. A symbol-only comparison detects 554 of the 555 self effects and misses QNG1→C9orf64, whose displayed names map to `ENSG00000165118`.

A sign reversal means that target abundance rose after knockdown in one state and fell after knockdown in the other. The census manifest stores the two orientation labels as `lost_on_activation` and `gained_on_activation`. A plug-in independent-normal simulation centred on the 79,110 observed concordant edge estimates and using their reported standard errors yielded a mean of 0.103 opposite-sign calls per realisation and 0.014 in the high-magnitude core, against a closed-form expectation of 0.0923. The operative catalogue definition remains significance in both endpoints plus opposite sign; the simulation is a plug-in sensitivity, not an empirical-FDR estimate.

## M13. Separate-dataset recurrence and cross-context structural comparison

Arce perturbations were estimated against the nine non-targeting guides in the reprocessed Perturb-CITE-seq Teff pseudobulks. Catalogue edges were joined to the Arce estimates by regulator and stable Ensembl target identity and required an Arce point estimate in both states; a regulator stratum contributed to the Mantel–Haenszel contrast only when it contained both catalogue-reversal and catalogue-stable edges. Leave-one-regulator-out estimates and within-regulator magnitude matching were sensitivities. Because the 28-regulator library was selected for state-specific factors and came from the same laboratory as the atlas, this estimates a within-regulator recurrence contrast among covered edges.

For the Song GSE249595 sensitivity, the state-specific joint ridge models were fitted over binary gene-carrier assignments and the targeted 374-gene expression panel. Catalogue edges were fixed before Song coefficients were joined and were compared with same-regulator stable edges; recurrence was the presence of opposite Song coefficient signs. The pre-specified manipulation-support gate required at least ten measurable focal self-transcripts and at least 70% of them to have negative coefficients in both states. Only ELAVL1 and NDUFA8 were measurable, although both were negative in both states. We therefore retained aggregate edge-level proportions and regulator-level values as descriptive cross-system sensitivity estimates and withheld the planned confidence interval and inferential tests.

For the independent Southard Hs27 and RPE-1 CRISPRa atlas [@southard2025comprehensive], guide and target identities were joined by exact versionless Ensembl identifiers. Guides had to belong to the final TF library, pass the released good-guide flags in both cell types, pass the seed-effect filters, have finite cell counts of at least 20 in each cell type and provide finite coefficients in both matrices. Context-specific regulator-target effects were medians over the same eligible guides. The primary test asked whether catalogue reversal dyads switched sign between Hs27 and RPE-1 more often than same-regulator stable dyads. Each reversal was matched without replacement to as many as five stable dyads using Rest and 48 h target expression, catalogue weaker-endpoint magnitude, Hs27 and RPE-1 target expression, Southard weaker-context magnitude and exact equality of common finite-guide count. All matching quantities were sign-blind and the match table was frozen before switch signs were summarized. The regulator-balanced risk difference was tested with 1,000,000 permutations of the reversal label within matched sets (seed 2026090401); adequacy required at least 50 mapped reversals before matching, 50 matched reversal sets and ten contributing regulators.

A separate Southard network was defined without CoDEG status. A high-confidence switch required at least two common dual-clean guides, opposite non-zero median signs, at least two adjusted-$p < 0.05$ guides supporting the median direction in each cell type, at least two-thirds guide-sign agreement in each cell type and no adjusted-$p < 0.05$ guide opposing the median; exact self pairs were excluded. Target convergence was $\sum_t \binom{k_t}{2}$. Within each regulator, the observed number of switch edges was reallocated without replacement among callable targets in joint quintile strata of target expression and sign-blind coefficient callability, preserving regulator switch degree and stratum composition (1,000,000 draws; seed 2026090402). Single orientation was tested among targets with switch in-degree at least three by permuting orientations among targets within regulator (1,000,000 draws; seed 2026090403). The two network tests formed one Holm-corrected family. Convergence required at least ten regulators and 50 switch edges; single orientation required at least ten eligible targets. No individual target, regulator or edge received a test.

## M14. Post-hoc regulator direction purity: exact null distribution

The eligibility set and the reversal graph were fixed. Within each target, the null treated all allocations of that target's observed lost and gained labels over its fixed incident stable-ID reversal dyads as equiprobable, and target blocks were independent. This fixes every regulator's reversal burden and every target's reversal in-degree and lost/gained totals.

The count distribution was evaluated exactly rather than by Monte Carlo or an independence approximation. For a subset $A$ of oriented regulator-purity events, its joint probability is $\prod_t (L_t)_{[a_t]}(G_t)_{[b_t]}/(n_t)_{[a_t+b_t]}$, where $L_t$, $G_t$ and $n_t$ are the target's lost, gained and total reversal counts, $a_t$ and $b_t$ are the numbers of selected events requiring each orientation, and $(x)_{[k]}$ is a falling factorial. Summing this probability over every subset of size $j$ gives $S_j = E\{\binom{N}{j}\}$; binomial-moment inversion gives the exact probability mass function of the number $N$ of pure regulators and the reported one-sided tail $P_{null}(N \geq N_{observed})$.

The four- through seven-reversal thresholds and their outcomes had already been inspected before this reproducibility implementation, so they are reported in full as a post-hoc threshold sensitivity. Code, exact rational outputs and their custody manifest are in `regulator_direction_purity_posthoc.py`, `regulator_direction_purity_posthoc.json` and `regulator_direction_purity_posthoc.manifest.json`.

## M15. High-magnitude reversal subnetwork: display construction

Regulator marker area was a monotonic square-root rescaling of out-degree within this restricted graph, and target marker area was similarly rescaled from in-degree. CCNC, TADA2B, SUPT7L, AHR and ARNT were labelled only to cross-reference panels e and f. Circular positions and curved bundles were deterministic layout devices, and the exploratory state-occupancy outlines in the full-size Repository Figure R5 were omitted.

## M16. Reversal-burden heterogeneity: tail models, dispersion models and covariate closure

The discrete power-law fit to regulator out-degree used the exact discrete Kolmogorov–Smirnov statistic, evaluated around each tied empirical jump, over an exhaustively searched $k_{min}$ retaining at least 50 tail observations (`degree_distribution_test.py`). Discretised lognormal, shifted geometric and exponentially cut off power-law tails were fitted on the same tail and compared by AICc and by non-nested Vuong tests. A tail-conditioned Poisson, the degree distribution of an Erdős–Rényi graph, was fitted separately (`degree_poisson_fit.py`) and compared to the power law on the same tail; it is reported outside the six-comparison set so that set is unchanged. The size-matched bipartite Erdős–Rényi comparator was fitted by the identical procedure (`degree_random_comparator.py`). The fit describes the observed census range.

Binomial overdispersion is by contrast estimable here, because it conditions on each regulator's own denominator: the gate determines which $n_i$ are observed, not the relation between $k_i$ and $n_i$ that the test evaluates. Departure from a constant per-edge reversal probability was tested by Tarone's Z and by a beta-binomial model in which logit $\mu_i = \beta_0 + \beta_1 \log_{10} n_i$, parameterised by the mean $\mu$ and the intra-class correlation $\rho$, with the degree slope and the dispersion each assessed by likelihood-ratio tests against the corresponding nested model. Robustness to the largest carriers was checked by refitting after removing the top 20 and 50 regulators by reversal count. The binomial interval taken around the fitted degree trend is conservative because it is discrete and because regulators share targets, so the target-margin null rather than the nominal 5% rate supplies the expected count.

Dispersion was additionally refitted with the regulator's median edge standard error, its on-target knockdown log2 fold change from the excluded self edges, a cubic degree term, and, on the subset with per-cell unit data, its measured cell support (`burden_covariate_closure.py`). The residual covers unmeasured technical sources such as guide-specific off-target activity and library representation, and is therefore not attributed to biology.

Outputs are `burden_heterogeneity.json`, the target-margin null in `target_null.json`, Figure 1e, Supplementary Figure S1b and the standalone diagnostic `Fig_burden_heterogeneity`. The degree fits are `degree_distribution_test.*.json` and `degree_random_comparator.json`, shown as Figure 1f; the full set, covering target in-degree and reversals per regulator as well as regulator out-degree, is repository figure R6.

## M17. Target functional groups and shared-target zooms: counting rules and code paths

All 2,075 reversal edges from the 49-regulator Figure 3c family were retained and ranked by total reversal count. No KEGG Human Diseases aggregate entered the dictionary. An edge was counted once in every group it matched; an edge matching none was counted once in the outside-six complement, so the group columns are non-additive. The panel reports raw lost and gained edge-count bins only. The route comparison quoted in the Discussion, covering the reversal-status partition over the 281 targets comparable for all three module regulators, the rank agreement of their net coefficients at each state and their on-target knockdown, is produced by `module_route_partition.py` and written to `module_route_partition.json`. The census counts quoted directly in the text are produced by `census_text_claims.py` and written to `manuscript_text_claims.json`.

In Figure 3e and f, Rest and 48 h coefficients use a common zero-centred colour scale. All three perturbations in the CCNC–TADA2B–SUPT7L intersection and both perturbations in the AHR–ARNT intersection have the same reversal orientation within each target.

## M18. Target-activation trajectory: expression processing, panel eligibility and coefficient movement

Ensembl version suffixes were removed from the GSE140244 matrix and duplicate gene rows were summed. When a donor/timepoint had technical A/B repeats, raw counts were summed before normalisation. We applied trimmed mean of M-values normalisation across the donor-time libraries [@robinson2010scaling] and calculated log2 counts per million with prior count 0.5. The submitted log2(TPM+1) matrix, analysed on the identical donor and target sets with technical repeats averaged within donor and time, was a pre-specified normalisation sensitivity.

The frozen unanimous display contains targets with at least two reversals running in one direction; unlike the specificity universe, its eligibility rule uses reversing edges only. In the per-donor Mann–Whitney probability, exact ties contributed one half. The nuisance-stratified sensitivity combined target baseline-expression quartile, census in-degree quartile and reversal-count stratum, retained strata containing both directions, and weighted their concordant and discordant target pairs. The stratification rule was frozen before the target join, but assignments to the baseline-expression quartile were then calculated from 0 h GSE140244 expression without using 48 h values. We repeated the display at minimum reversal counts of three and five. The count matrix itself had already been streamed for format, integer/non-negative and sample-alignment checks, so the accurate custody claim is target-outcome-blind and pre-target-join, not file-unopened. The signed-score question and the target-conditioned null were written before the GSE target join but executed after the unanimous-panel result had been inspected, and are therefore designated a prewritten post-primary specificity analysis.

As a retained post-hoc sensitivity reported in Supplementary Data 17 and 18, we mapped the same donor-median external response to 4,344 reversing and 78,520 stable trans-census edges and defined coefficient movement as $|\mathrm{lfc}_{48} - \mathrm{lfc}_{Rest}|$. Within each regulator, separately for reversing and stable edges, we calculated the raw Spearman correlation between target response and coefficient movement, requiring at least four mapped edges of each kind. We compared the correlations across 154 paired regulators with a two-sided paired Wilcoxon test and repeated the analysis at minimum edge-count floors of five, six, eight, ten and 12. At a floor of eight, a secondary partial-Spearman sensitivity rank-residualised both variables jointly on the Rest and 48 h standard errors.

## M19. Genome-scale state decomposition: activation-axis validation and outputs

A score trained on three donors separated the two conditions in the held-out fourth at a leave-one-donor-out Cliff's  of 0.69. The computation used a single 1.13 TB streaming pass with staged canary checks. Outputs are released as `atlas_composition_FINAL.json` and `atlas_edge_composition.csv.gz` and plotted in Supplementary Figure S2c,d.

## M20. Binary reversal-target KEGG sensitivity and network-adjusted all-target rank enrichment

The initial binary sensitivity tested whether the 2,577 targets carrying at least one reversal were over-represented among all 9,554 eligible census targets in the pinned KEGG Legacy v2026.1 collection. Human-disease and drug-development categories were excluded, and terms with 15–200 eligible targets were retained, giving a fixed family of 113 pathways. Conventional one-sided Fisher exact p values were Benjamini–Hochberg adjusted within this family. Because targets differed in their number of eligible incoming edges, the primary calibration sampled each regulator's observed reversal count without replacement from that regulator's complete eligible trans-target pool, collapsed the sampled edges to unique targets and recomputed the maximum hypergeometric-standardised overlap across all 113 terms in each of 20,000 permutations. This preserved every regulator's reversal burden and eligible target pool. Separately, 20,000 permutations reassigned the 2,577 binary reversal-target labels within each exact census in-degree stratum, preserving target in-degree and total query size. The binary analysis was interpreted as descriptive unless the complete family passed the global maximum-statistic gate and an individual term passed single-step maxT correction.

The signed-orientation display used the same fixed KEGG family but a separate estimand. For each informative target, the target-conditioned null fixed its total eligible in-degree, resting-sign composition and reversal count; the centred contribution was the observed lost-reversal count minus its exact hypergeometric expectation. Pathway contributions were summed and variance-standardised, with positive values denoting excess lost orientation and negative values excess gained orientation. A pathway required at least ten informative targets. The complementary regulator-margin sensitivity permuted lost/gained labels across the fixed reversal edges within each regulator, preserving every regulator's orientation burden. Each null was separately centred and scaled before two-sided single-step maxT correction over the 60 eligible signed pathways, using 20,000 draws. The two p values were not combined: passing both tests does not condition target and regulator margins simultaneously and supports orientation over-representation only, not pathway activity or mechanism.

This post-hoc exploratory method correction asks whether genes in an annotated term tend to carry unusually high pooled reversal burden after allowing for their opportunities to reverse. The universe contains all 9,554 targets incident to at least one of the 83,489 eligible trans edges, including 6,977 targets with no observed reversal. For each target, we summed its observed reversals across the 1,234 regulator-by-Rest-sign strata. Within each stratum, the exact null expectation and variance follow from sampling without replacement the observed number of reversals from that stratum's eligible edges. The target score is the observed pooled reversal count minus this exact expectation, divided by the exact null standard deviation. The 229 targets with zero exact null variance remain in the universe with score zero. Targets were ordered by decreasing score and then by the frozen stable-ID index.

We tested KEGG Legacy v2026.1 and GO Biological Process 2023 gene sets after exact human-symbol mapping to this universe. Terms with 10–300 mapped targets were retained, giving 168 KEGG and 2,425 GO terms. For each term, a one-sided weighted running-sum statistic tested whether its genes accumulated near the high-burden end of the ranking. The displayed network-null-normalised enrichment score is the observed enrichment score divided by that term's mean permutation enrichment score. Inference used the exchangeably standardised score, not a parametric approximation. The stored leading edge includes term members through the positive peak; a second field expands it across all target-score ties at that peak.

Two independent banks of 10,000 network permutations used seeds 2026090221 and 2026090222. Every permutation sampled without replacement within each regulator-by-Rest-sign stratum, preserving its reversal count and eligible target pool and therefore preserving the census totals of 2,298 lost and 2,081 gained reversals. One-sided plus-one marginal $p$ values were Benjamini–Hochberg adjusted separately within the KEGG and GO families. A term was called statement-eligible only if its within-family $q$ value was below 0.10 in the pooled 20,000 draws and in each bank separately. A stricter secondary test used synchronized single-step maxT correction across all 2,593 terms; pooled $p < 0.05$ denotes global support, and the same result in each bank denotes global stability.

Null calibration was independent across banks: rows from bank A were tested only against bank B, and vice versa. The independent audit then regenerated every graph and all 2,593 enrichment scores per draw through a separate flat-membership implementation that did not import the production module. It replayed all 10,000 draws in each bank, verified the released inference from the checksum-bound stored scores and confirmed that floating-point roundoff changed no scientific flag. The analysis and audit artifacts are in `results_data/census/broad_alltarget_kegg_go_rank_enrichment_v2`; Supplementary Data 19–21 provide the complete term table, target scores and leading-edge memberships.